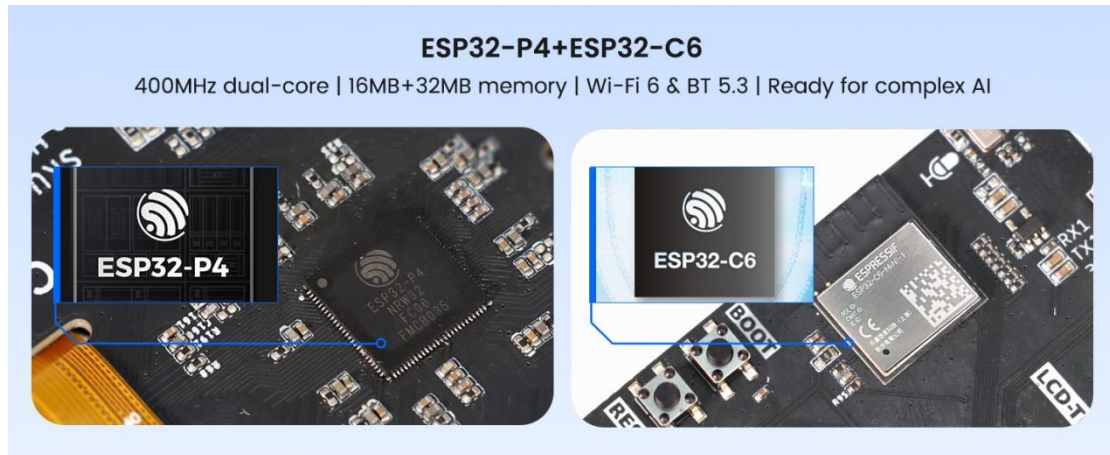


CrowPanel Advance ESP32-P4 Display

Firmware Upgrade Guide



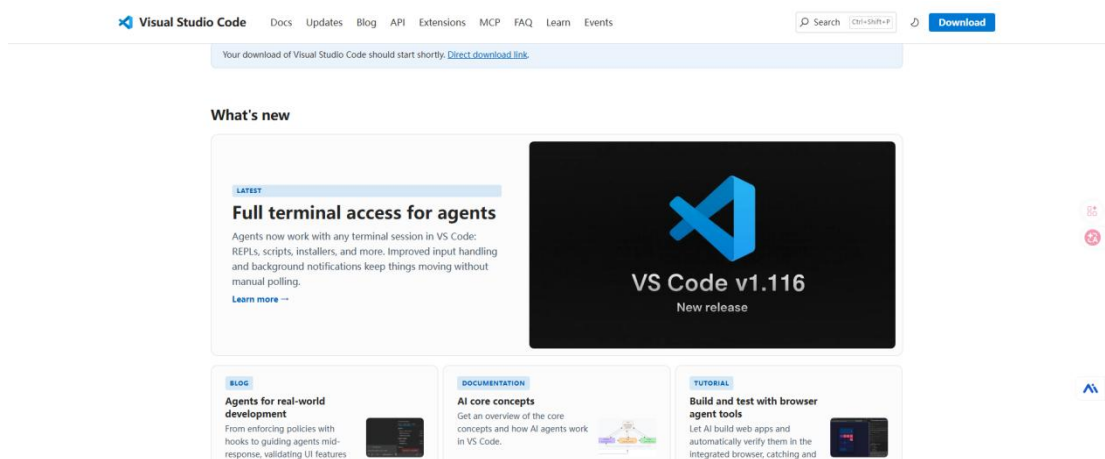
Compile and flash using the PlatformIO plugin on VS Code.

This tutorial will guide you through the firmware upgrade operation of the CrowPanel Advance ESP32-P4 HMI development board. The main steps include: installing the VS Code development environment, installing the PlatformIO plugin, installing Git and cloning the project code, and finally compiling and flashing the firmware to the device through PlatformIO to realize the synchronous firmware upgrade of the ESP32-C6 chip and ESP32-P4 chip.

1. Download and Install VS Code

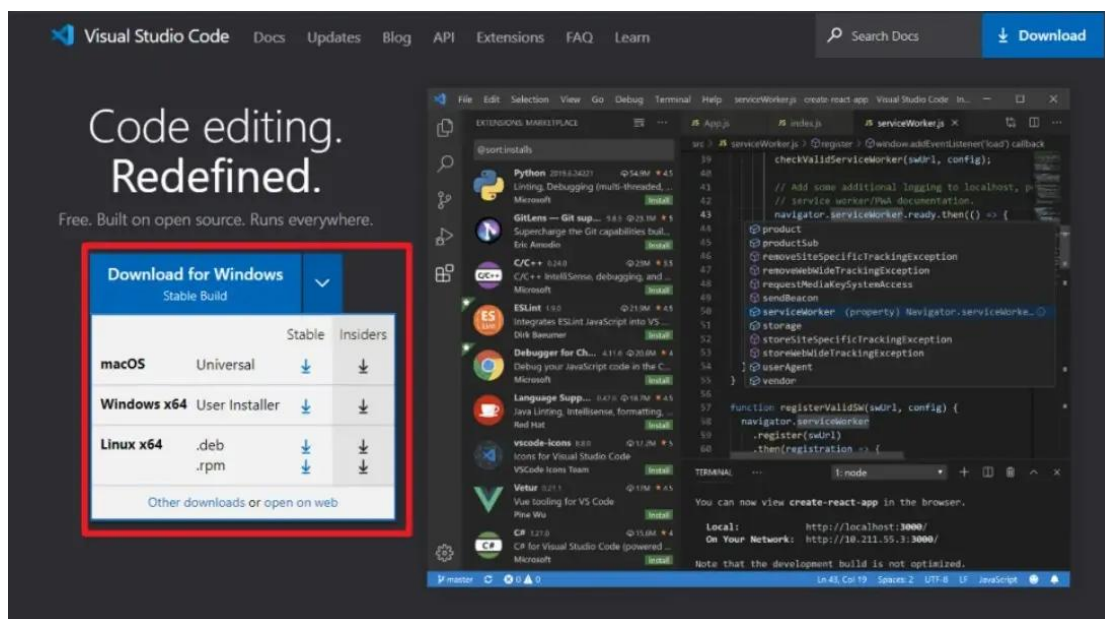
VS Code (Visual Studio Code) is a lightweight code editor developed by Microsoft, supporting multiple programming languages and plugin extensions. It is the recommended host tool for the PlatformIO development environment. Please follow the steps below to download and install it.

<https://code.visualstudio.com/thank-you?dv=win64user>



VS Code Installation

(1) First, download Visual Studio Code from <https://code.visualstudio.com/>. Select the version compatible with your computer's operating system and download it.

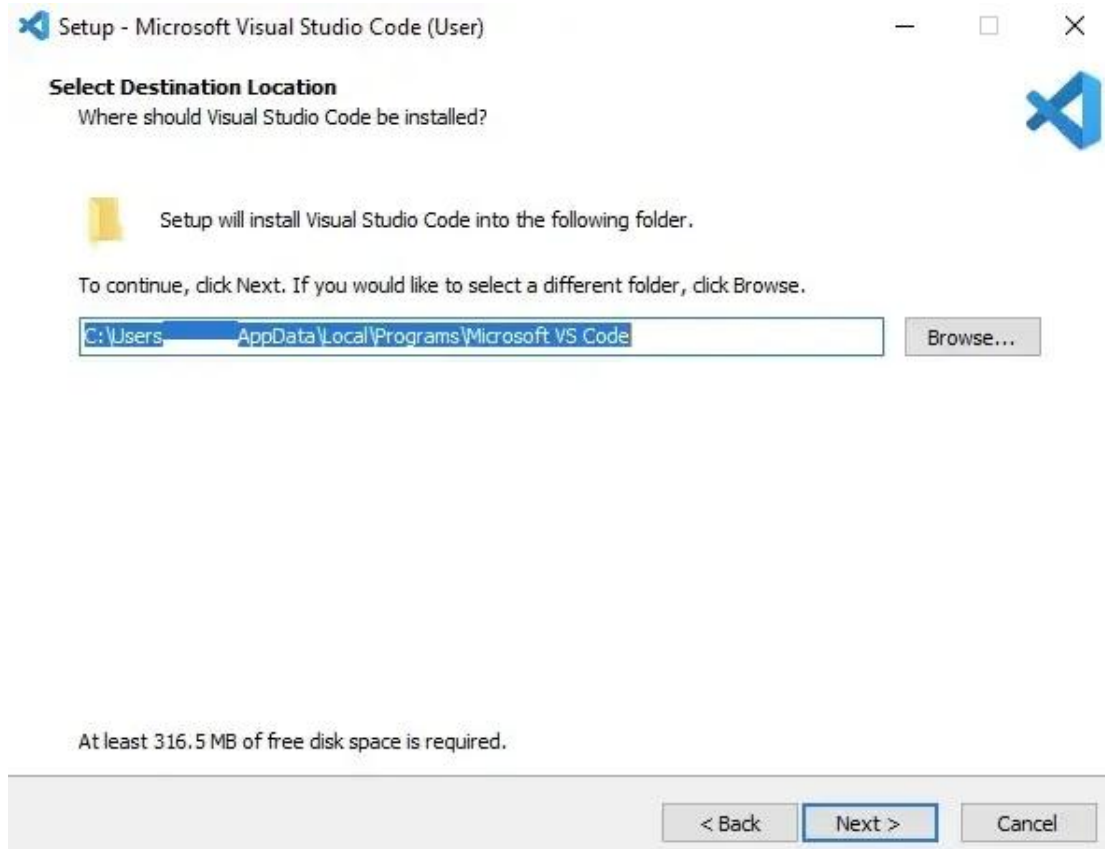


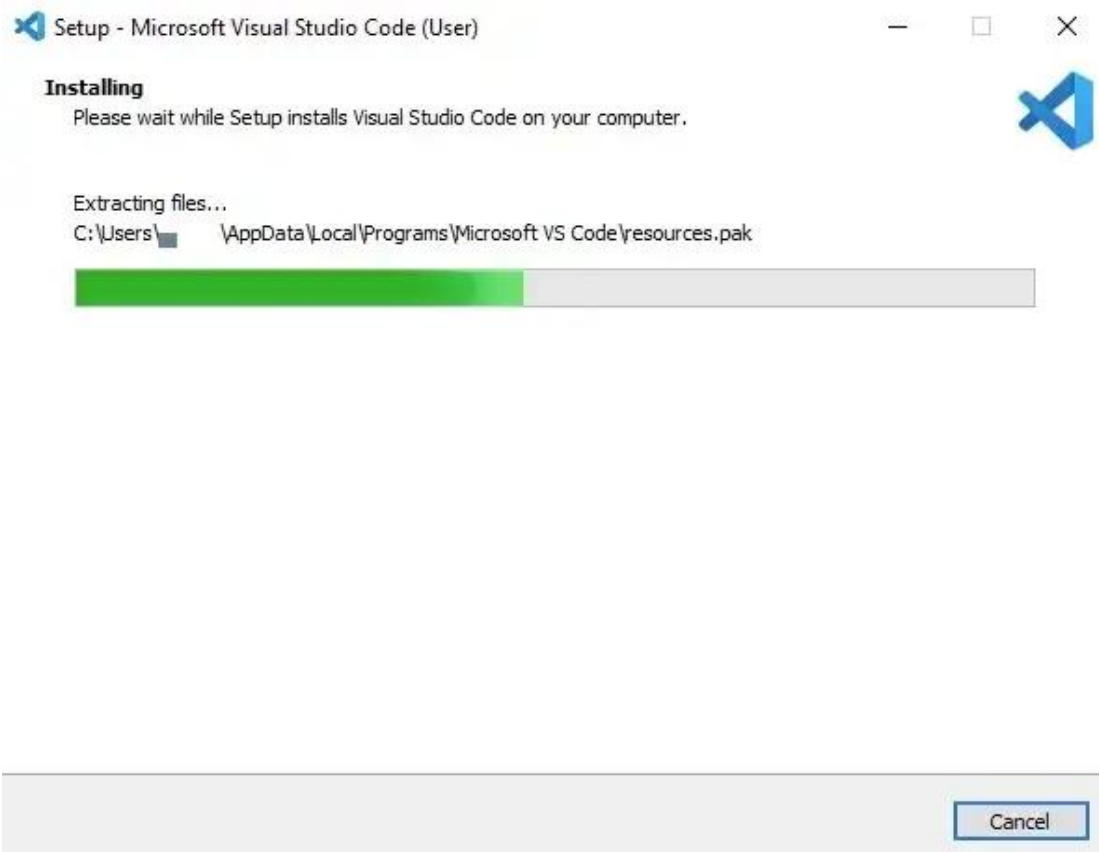
(2) Double-click to install the Visual Studio Code software, then use the default settings for the entire installation process.

Check the box and click "Next". The following screenshot is an example of the installation process:



Select the installation path.





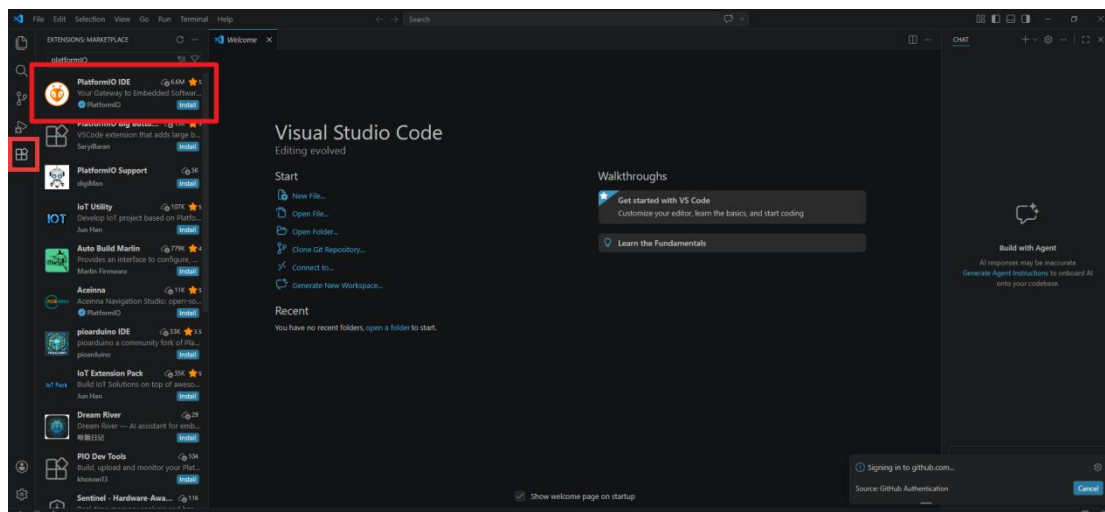
Click "Finish" after the installation is complete. If "Launch Visual Studio Code" is checked, VS Code will start automatically.



Open VS Code and install the PlatformIO IDE plugin.

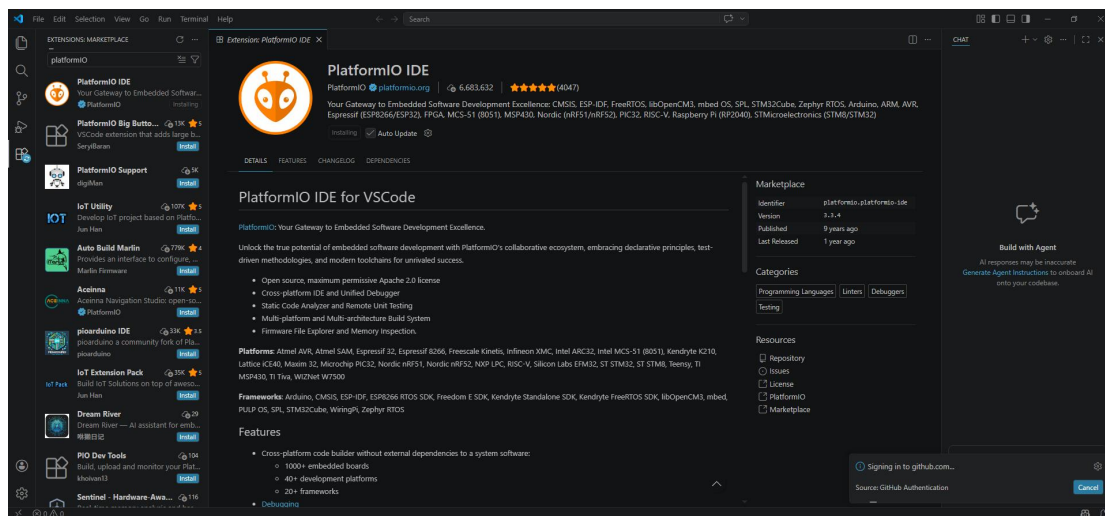
PlatformIO is a cross-platform development environment designed specifically for embedded development, supporting various microcontrollers such as ESP32. Please follow the steps below to install it:

- (1) Open VS Code and click the "Extensions" icon in the left activity bar.
- (2) Enter "PlatformIO IDE" in the search box and find the corresponding plugin, as shown in the figure below:



(3) Click "Install" to install. After the PlatformIO IDE plugin is downloaded, it will automatically update the development environment. Please wait patiently (usually a few minutes, depending on network speed). After the installation is complete, the PlatformIO ant icon will appear in the status bar at the bottom of VS Code, indicating successful installation.

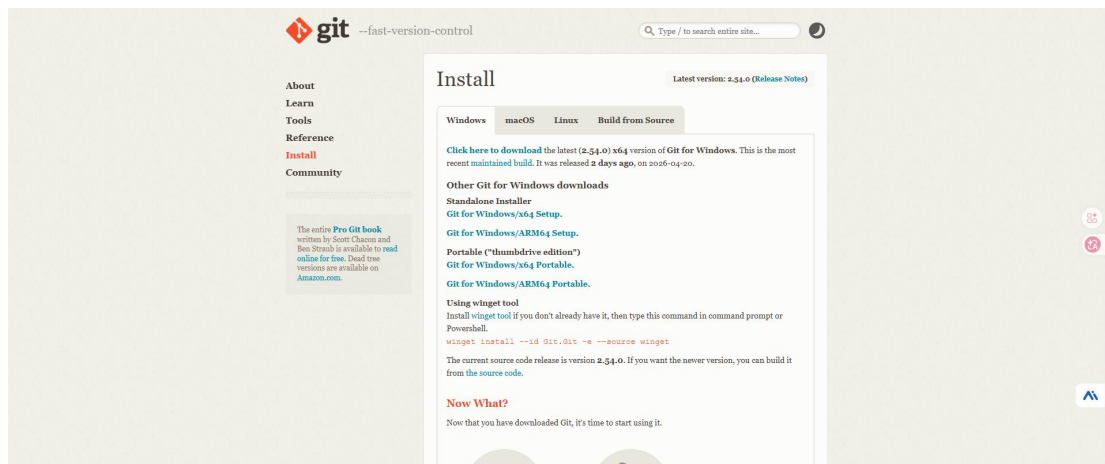
Note: The installation process requires an internet connection. Please ensure the network is stable. If the installation speed is slow, you can try using a network proxy or retry after a while. If prompted to reload (Reload Required) after installation, click "Reload Now" to restart VS Code to complete activation.



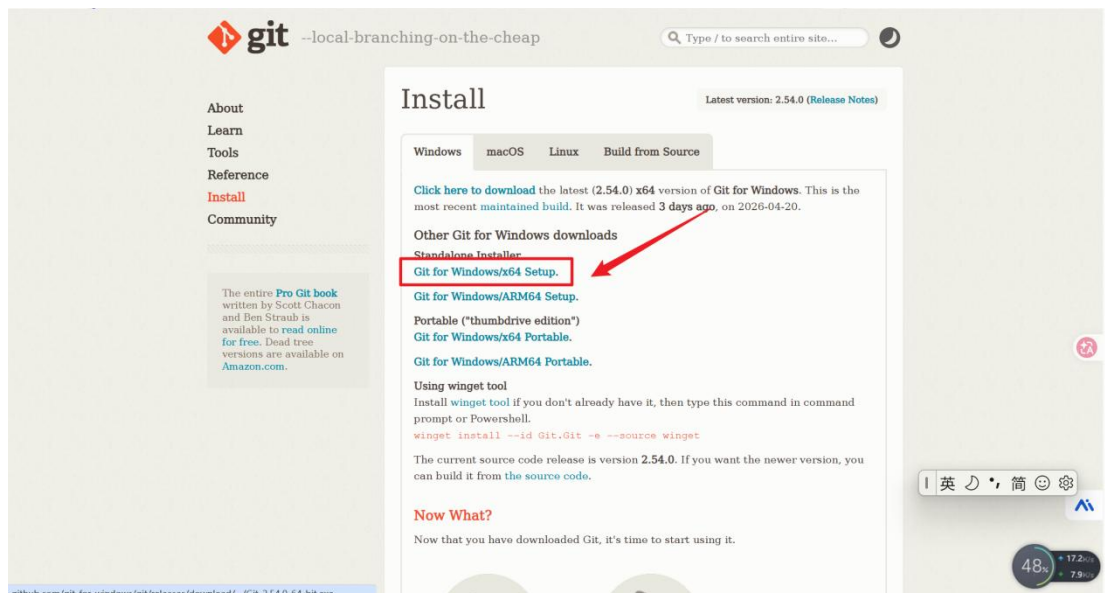
2. Download and Install Git

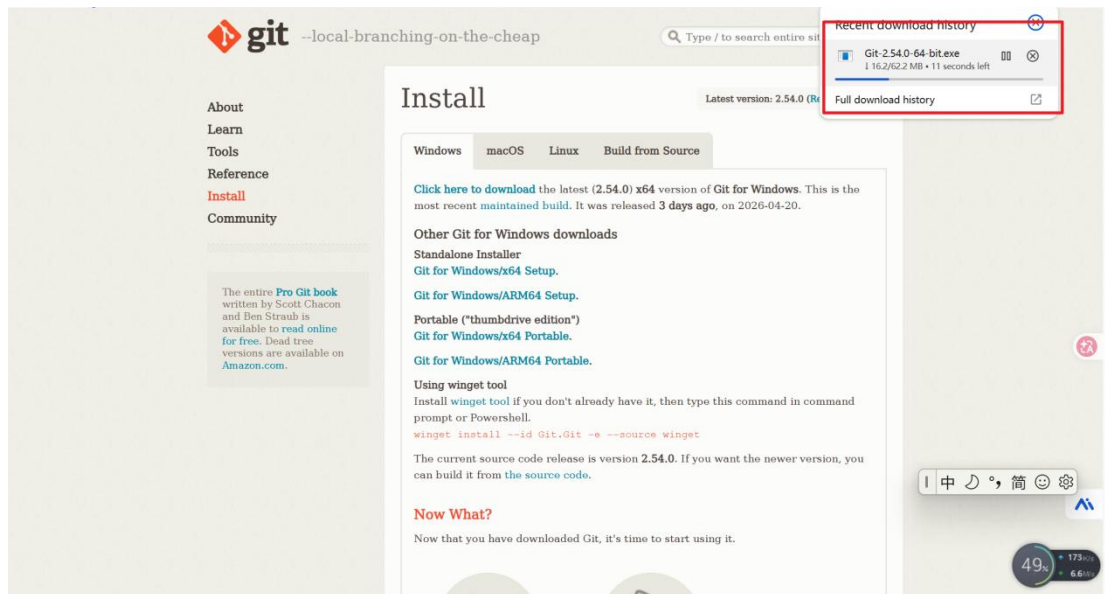
Git is a distributed version control system used to download project source code from GitHub. Please visit the following link to download the Git for Windows installation package:

<https://git-scm.com/install/windows>

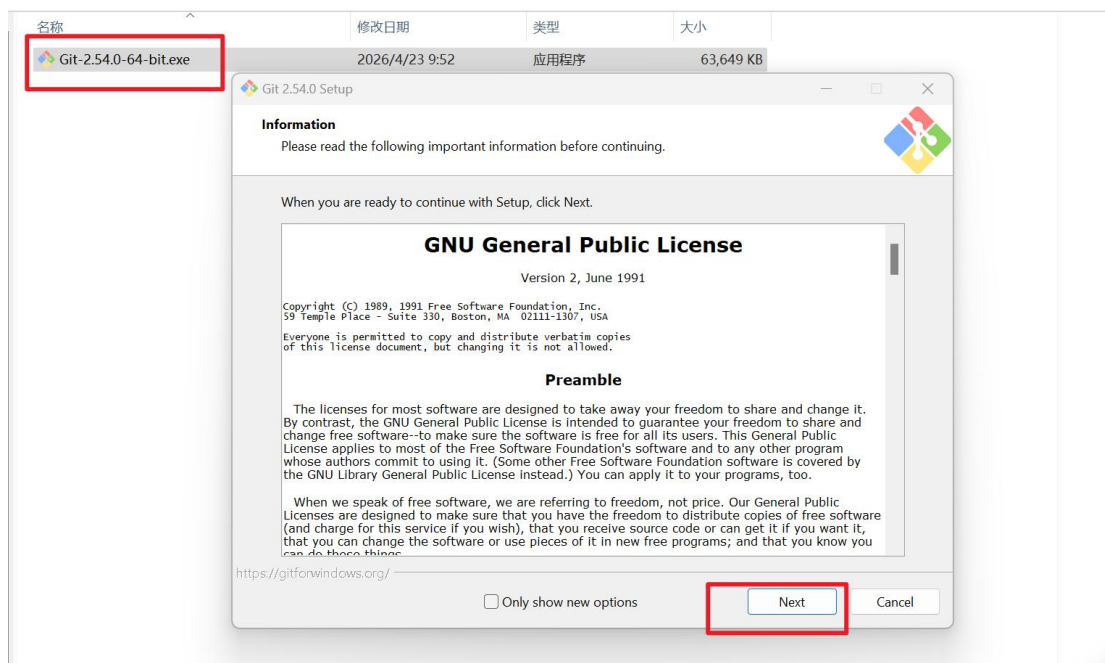


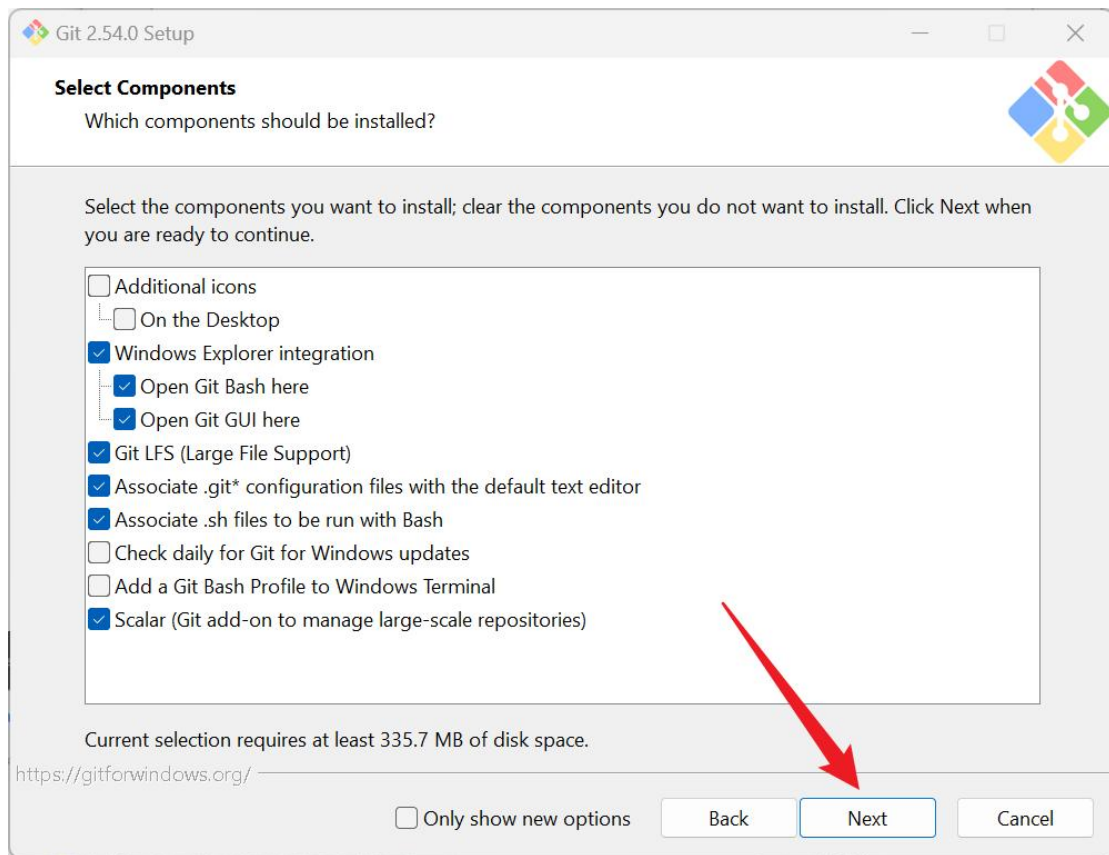
Select the version corresponding to your computer's operating system for installation; here we select Windows/x64. After the download is complete, double-click the installation package to start the installation wizard, as shown in the figure below:



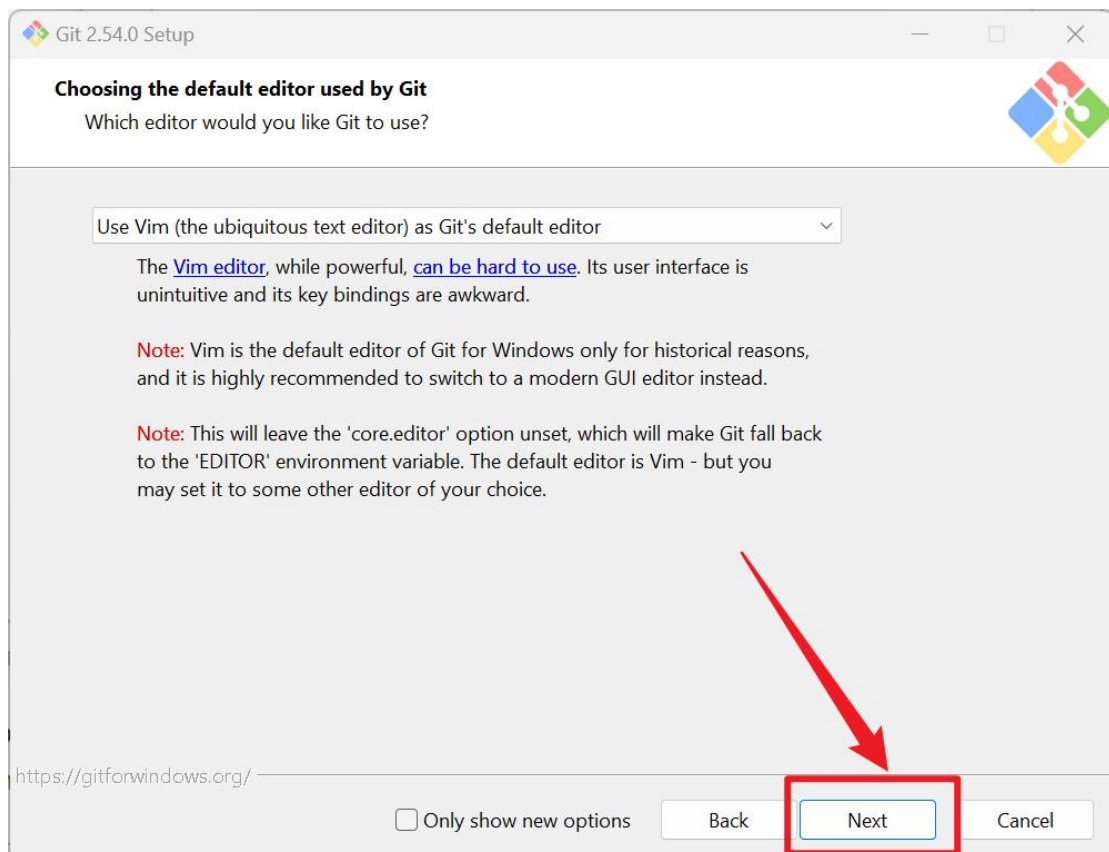


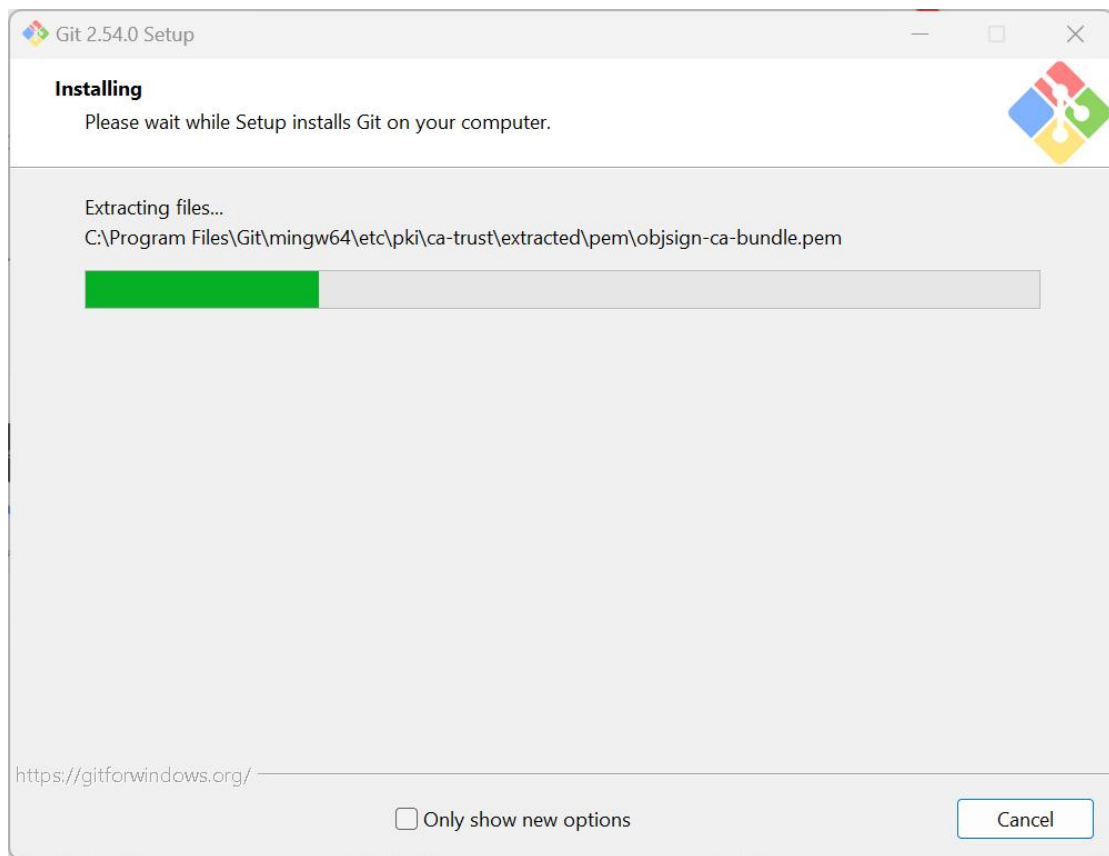
Open the downloaded Git installation file and click "Next". Keep the default options on each page during the installation process, and continue clicking "Next" until the installation is complete, as shown in the figure below:



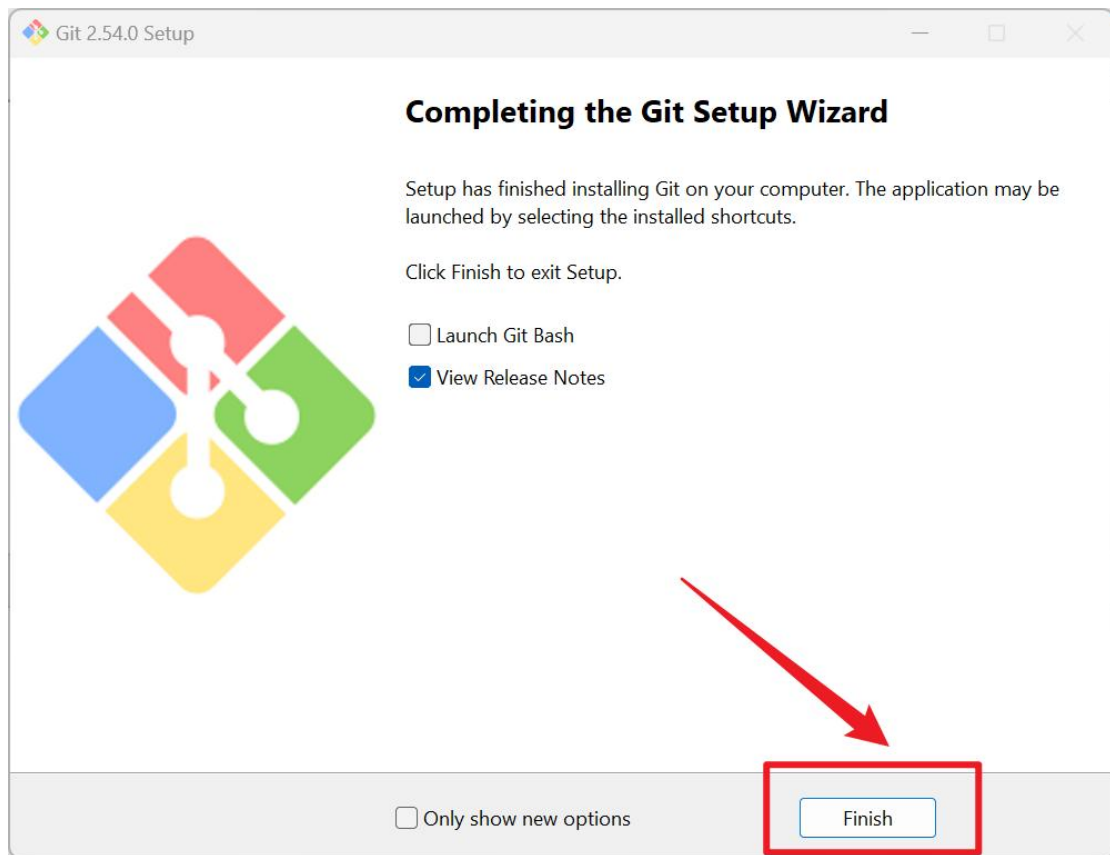


Keep the default options and continue clicking "Next";





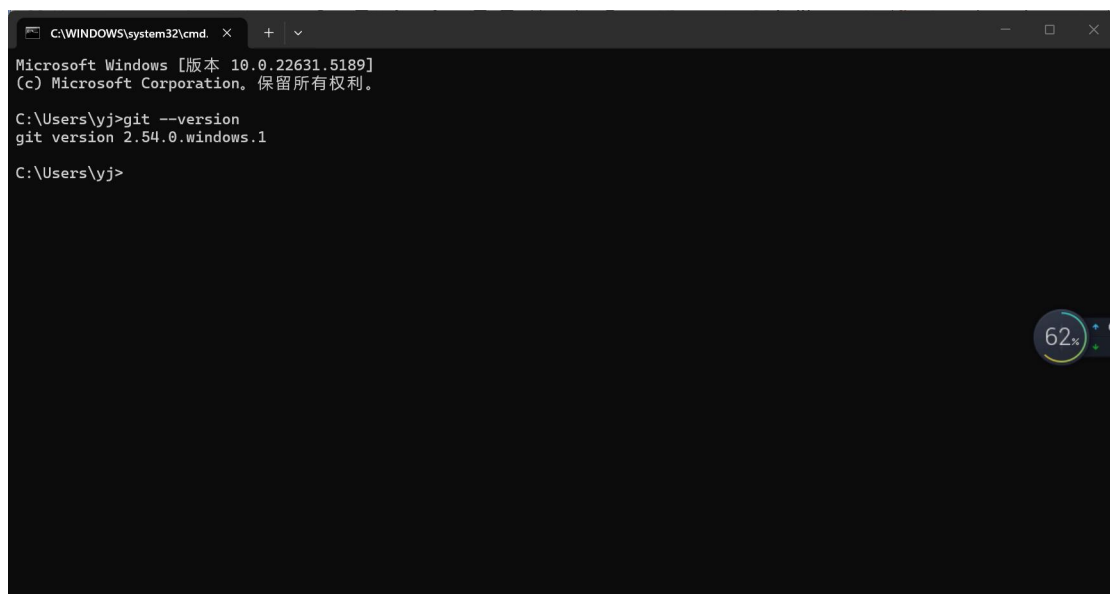
Until the installation is successful, as shown in the figure below, finally click "Finish" to complete the installation.



Verify installation: Open the Command Prompt (press Win+R, enter cmd and press Enter), and enter the following command:

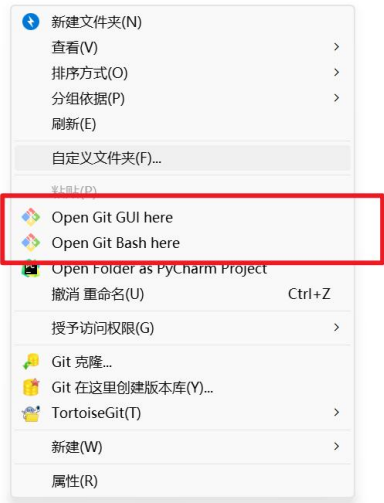
```
git --version
```

If the Git version number is displayed (e.g., git version 2.x.x), the installation is successful.



After successful installation, right-click in the blank space of any folder, and you will see two options: "Open Git GUI here" and "Open Git Bash here", indicating that Git has been correctly integrated into the system right-click menu, as shown in the figure below:

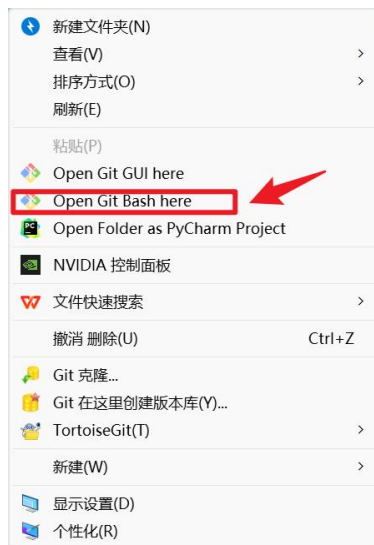
名称	修改日期	类型	大小
 Git-2.54.0-64-bit.exe	2026/4/23 9:52	应用程序	63,649 KB



3. Download Project Files Using Git Commands

The project source code is hosted on GitHub and needs to be cloned to the local computer using Git tools. The specific steps are as follows:

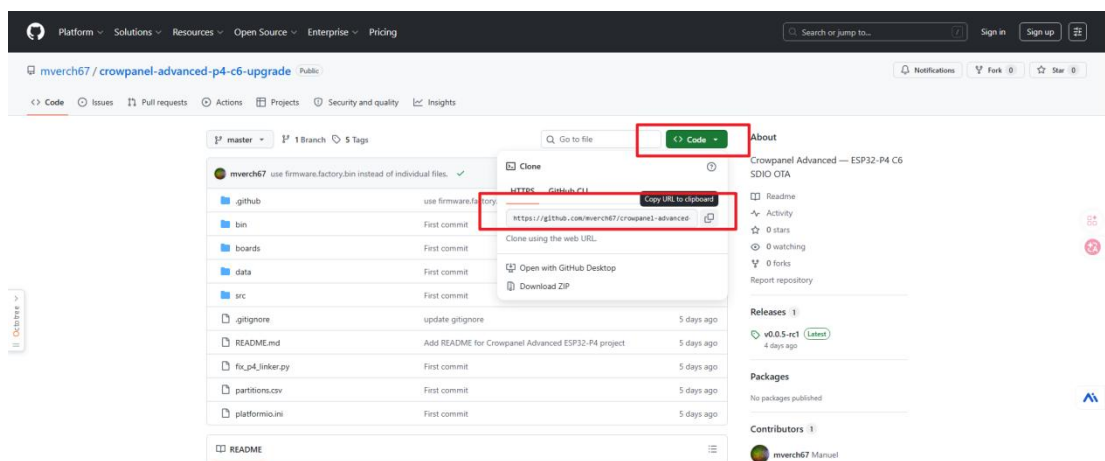
(1) Right-click in the blank space of the folder where you want to store the project (e.g., desktop or root directory of drive D) and select "Open Git Bash here" to open the Git Bash command line terminal.

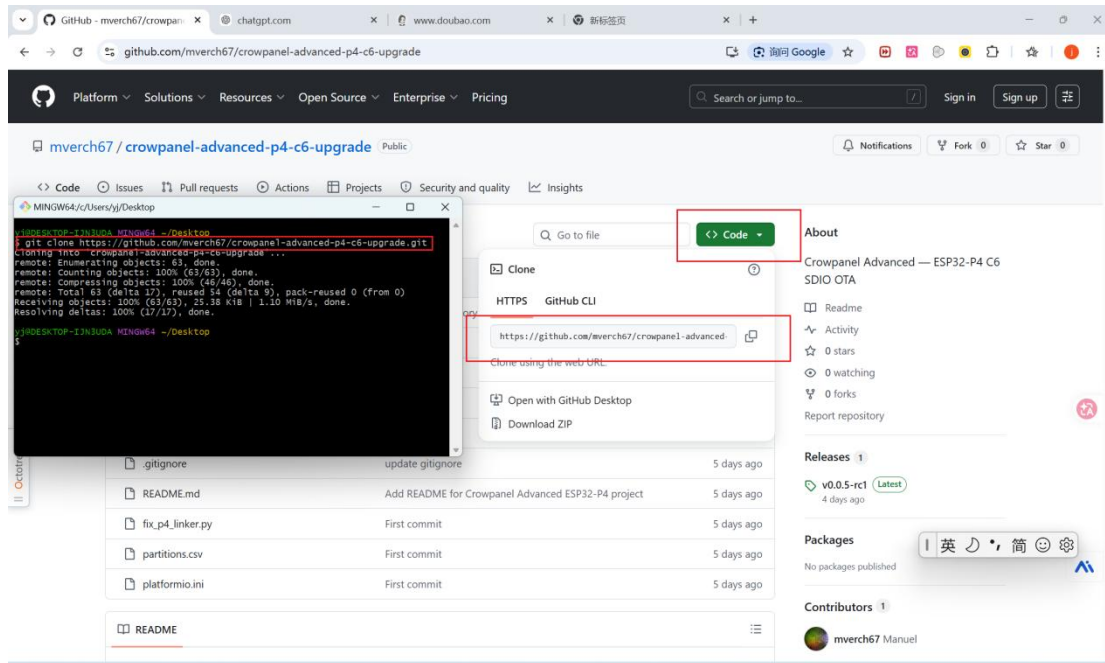


(2) Use the Git clone command to download the files. The following screenshot shows the process of executing the clone command in Git Bash:

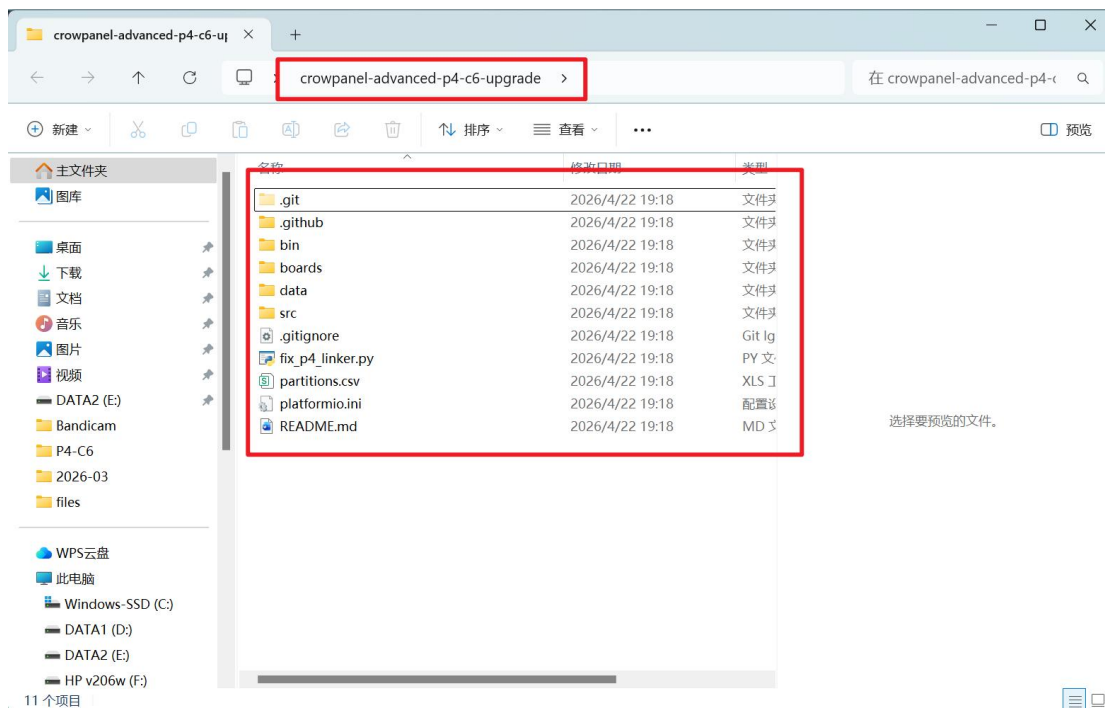
git clone <https://github.com/mverch67/crowpanel-advanced-p4-c6-upgrade>

Enter the following clone command in the Git Bash terminal and press Enter to execute:





The content of the downloaded project folder is shown in the figure below. The folder contains the source code files and configuration files required for the PlatformIO project.

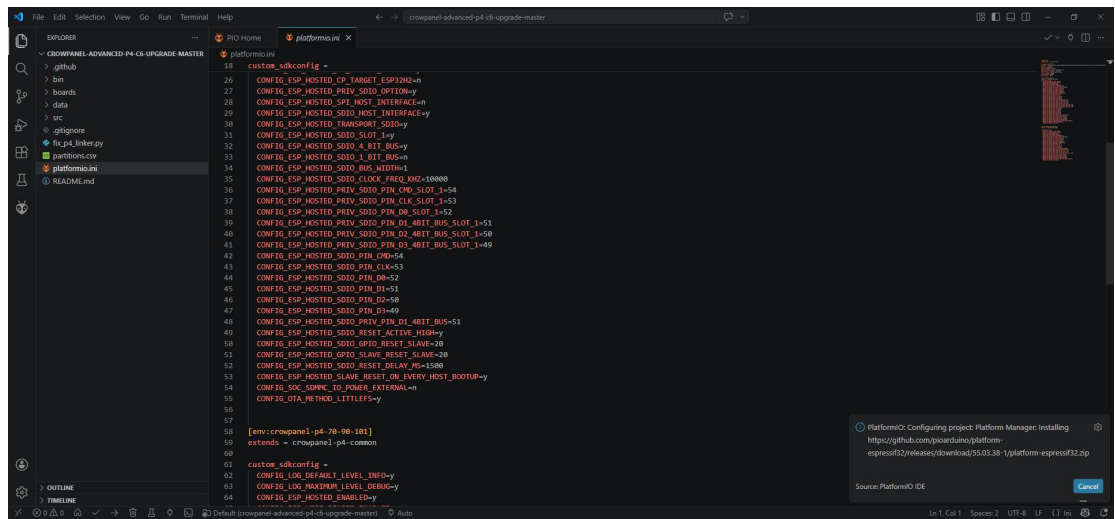


4. Open the Project in VS Code and Compile/Flash

Right-click on the downloaded project folder and select "Open with Code", or directly drag the project folder into the opened VS Code window.

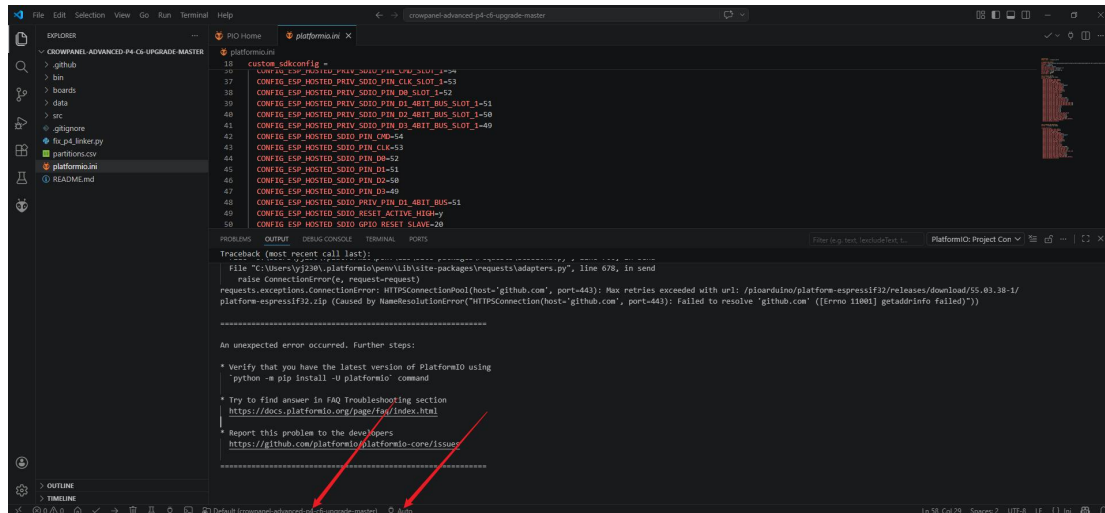


After opening the project folder in VS Code, PlatformIO will automatically recognize the project configuration and start loading the development environment (including toolchain, dependent libraries, etc.). Please wait patiently. The status bar at the bottom of VS Code will display the loading progress. After loading is complete, you can proceed to the next step, as shown in the figure below:



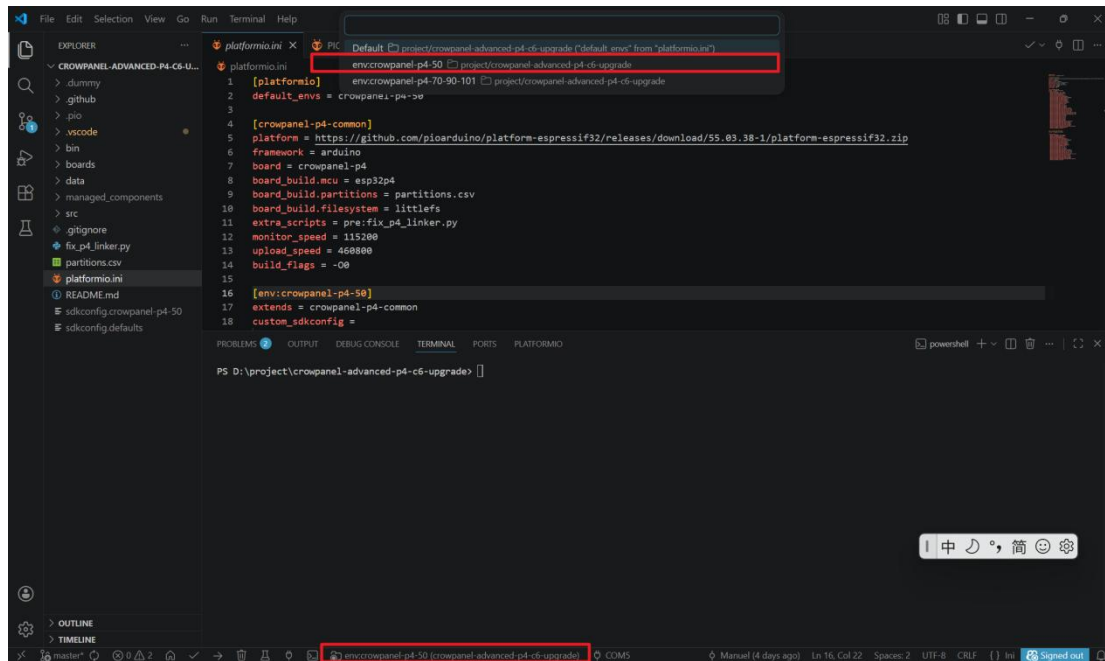
5. Select the Corresponding Device Model and Serial Port

Before flashing the firmware, you need to select the correct device model and serial port to ensure that the program can be correctly flashed to the target board.



Device selection for compilation environment: In this example, an ESP32-P4 5.0-inch board is used to flash the firmware. (Device selection for compilation environment: choose between 5-inch and 7-inch, 9-inch, 10.1-inch screens)

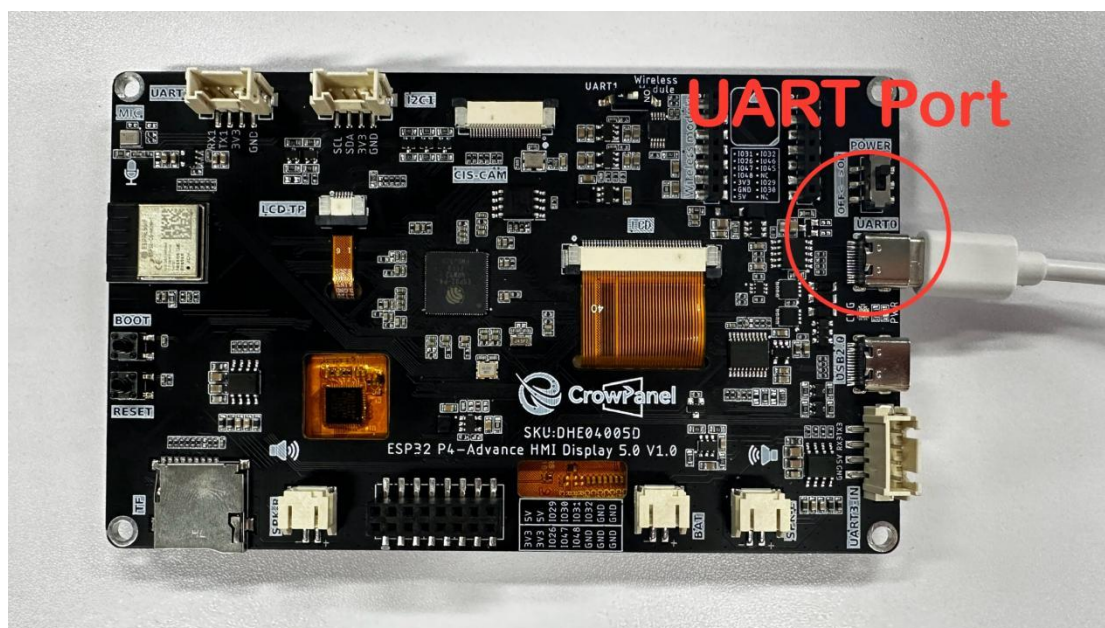
Click "Default (crowpanel-advanced-p4-c6-upgrade)" in the status bar at the bottom of VS Code to switch the device environment, and select the configuration corresponding to your hardware size, as shown in the figure below:

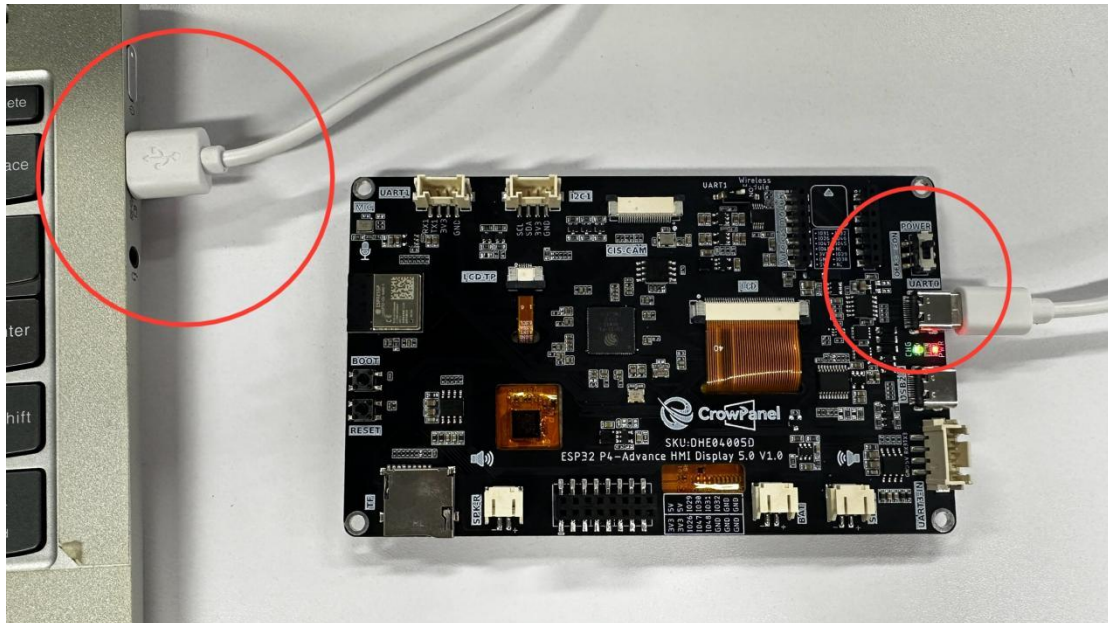


```
platformio.ini
env: Default [P] project\crowpanel-advanced-p4-c6-upgrade ("default: env" from "platformio.ini")
env:crowpanel-p4-50 [P] project\crowpanel-advanced-p4-c6-upgrade
1 [platformio] env:crowpanel-p4-70-90-101 project\crowpanel-advanced-p4-c6-upgrade
2 default_envs = crowpanel-p4-50
3
4 [crowpanel-p4-common]
5 platform = https://github.com/pioarduino/platform-esp8266/releases/download/55.03.38-1/platform-esp8266.zip
6 framework = arduino
7 board = crowpanel-p4
8 board_build.nuc = esp32p4
9 board_build.partitions = partitions.csv
10 board_build.filesystem = littlefs
11 extra_scripts = pre:fix_p4_linker.py
12 monitor_speed = 115200
13 upload_speed = 460800
14 build_flags = -O0
15
16 [env:crowpanel-p4-50]
17 extends = crowpanel-p4-common
18 custom_sdkconfig =
```

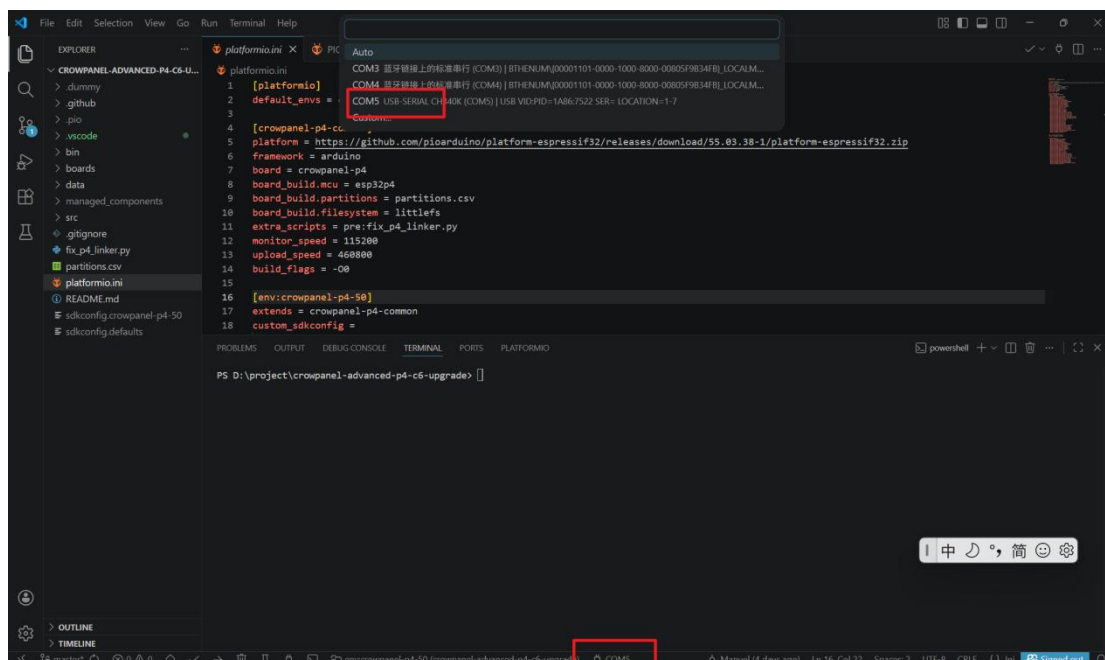
PS D:\project\crowpanel-advanced-p4-c6-upgrade>

Connect the UART interface (marked with UART) of the board to the USB interface of the computer using a USB-C data cable. Note: A USB-C cable that supports data transmission must be used. After connection, the computer will automatically install the driver program. The correct wiring method is shown in the figure below:





Serial port selection: After connecting the USB-C cable, click the serial port selection area in the status bar at the bottom of VS Code (displaying "Auto" or the last selected serial port), and select the serial port corresponding to the board from the pop-up list (usually COM3, COM4, etc. on Windows. If there are multiple serial ports, you can plug and unplug the USB cable to determine which one is the target serial port), as shown in the figure below:



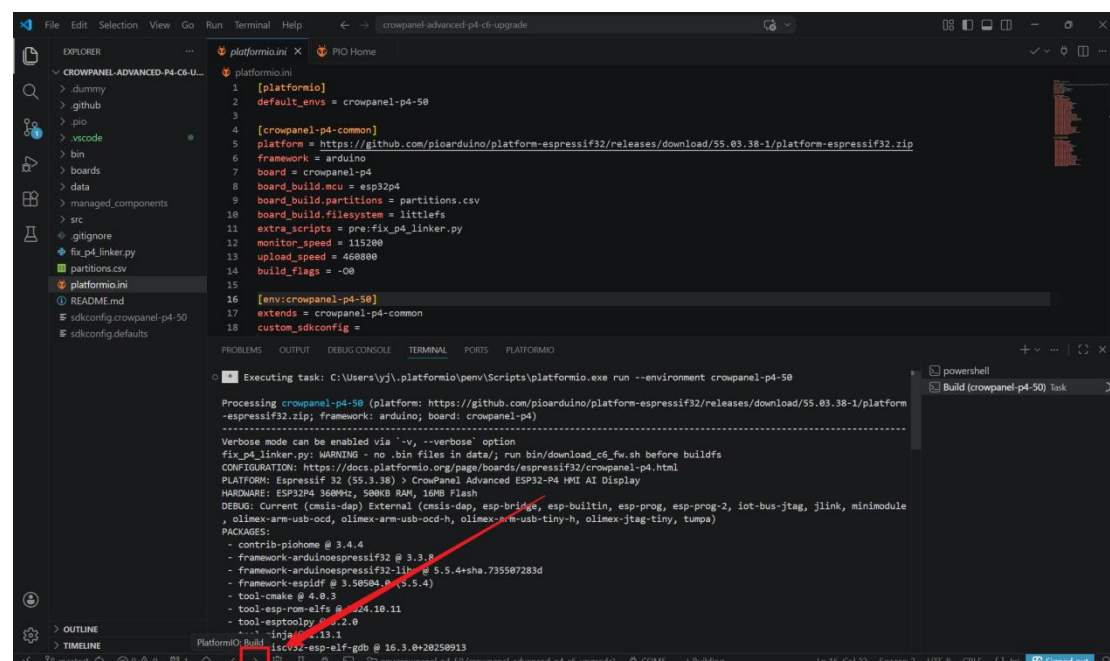
6. Compile and Upload the Firmware

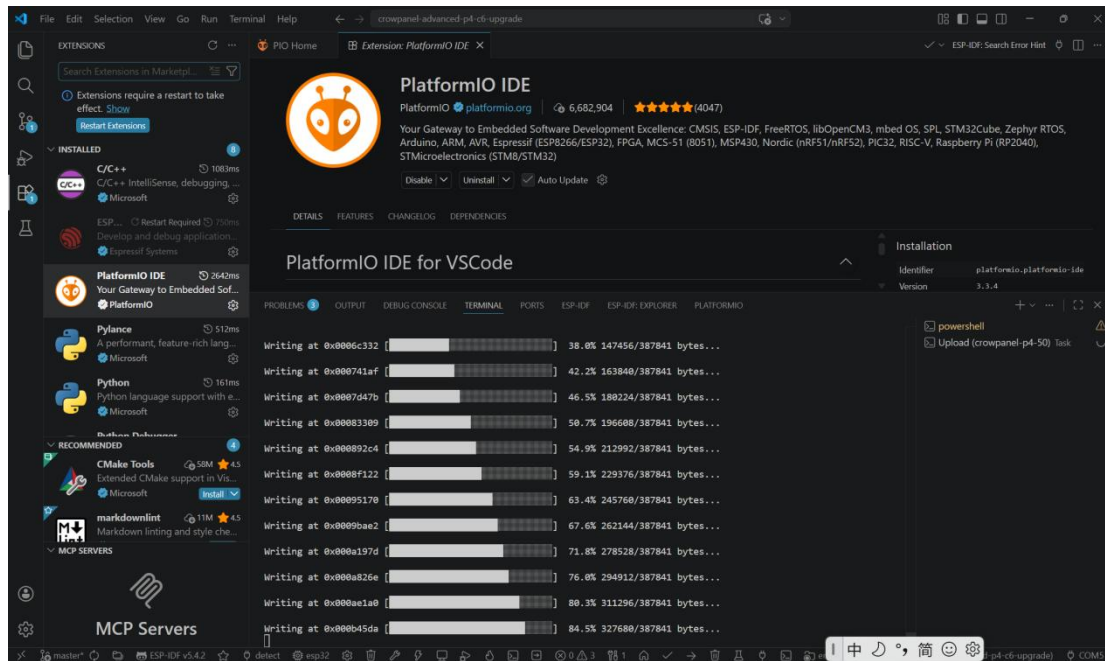
Click the "→" (Upload) button in the status bar at the bottom of VS Code. PlatformIO will automatically complete the following operations:

- Compile the source code to generate firmware (.bin file)
- Flash the firmware to the ESP32-P4 chip through the selected serial port

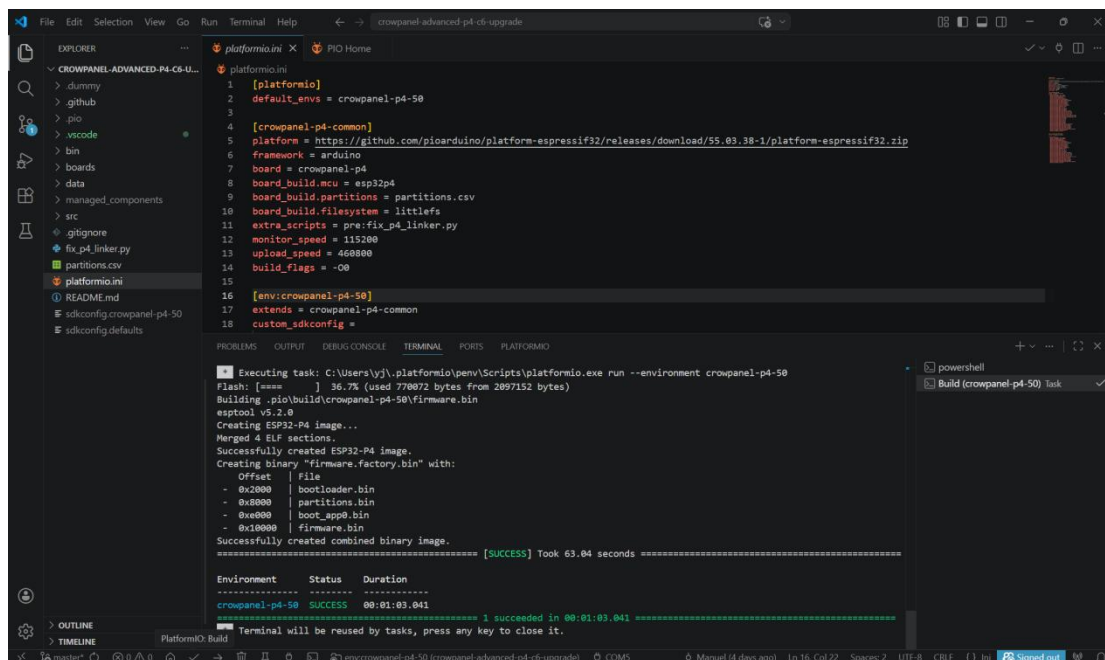
During the flashing process, the terminal panel at the bottom of VS Code will display real-time progress information. The operation interface for executing the upload is shown in the figure below:

Note: Clicking the "→" (Upload) button will first compile and then upload.

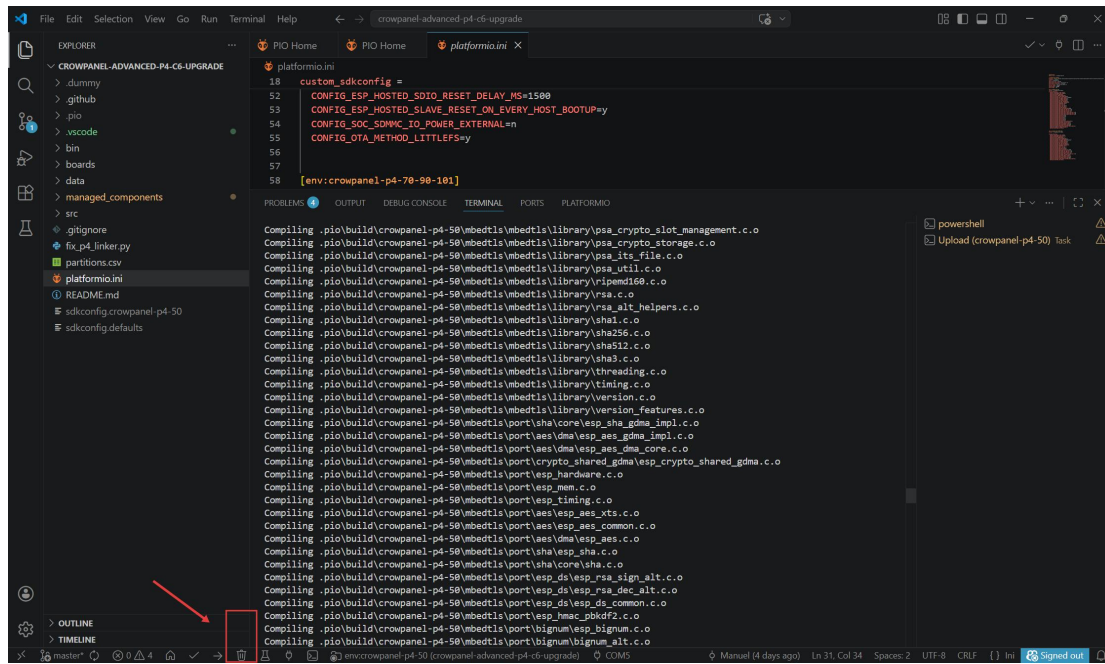




After successful flashing, the Status in the terminal panel will display "SUCCESS", indicating that the firmware has been successfully written to the device, as shown in the figure below:



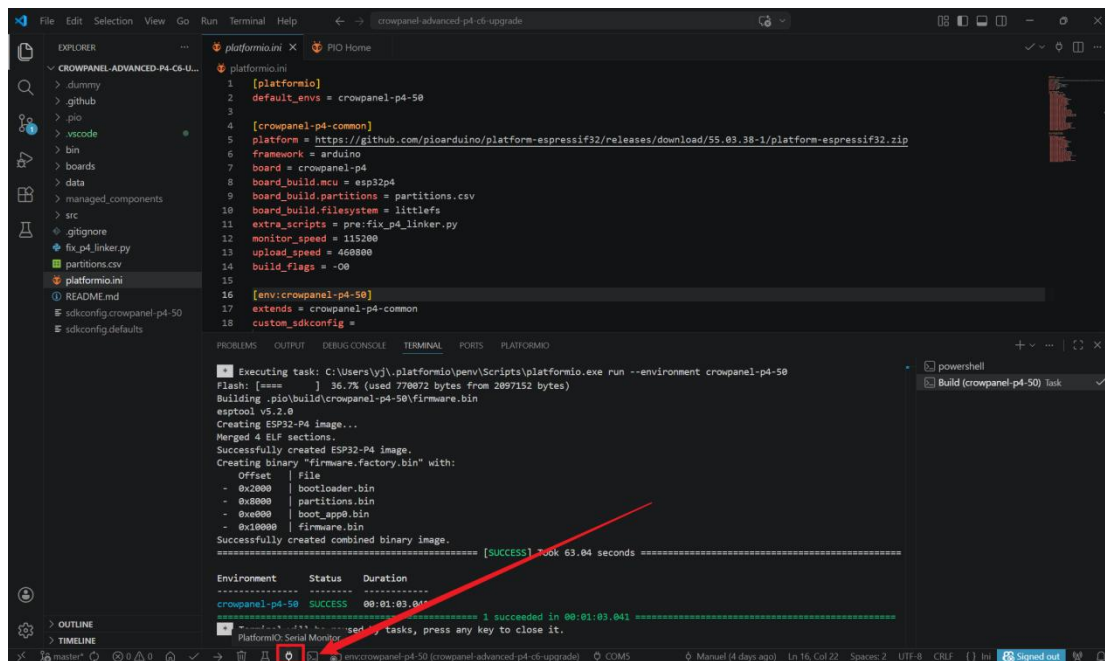
Note: If the compilation fails after switching the board, click the Clean button to clean up, then try compiling again.



7. View Serial Monitor Print Information

After the flashing is complete, click the serial monitor icon (plug icon) in the status bar at the bottom of VS Code, or press the shortcut key to open the serial monitor and view the print information output by the device.

The print information in the serial monitor will display the GPIO pin number and the ESP32-C6 firmware upgrade version information. If you see "OTA upgrade success" or a similar success prompt, it indicates that the firmware of the ESP32-C6 chip has been upgraded synchronously, as shown in the figure below:



```
platformio.ini
18 custom_sdkconfig =
36 CONFIG_ESP_HOSTED_PRIV_SDIO_PIN_CMD_SLOT_1=54
37 CONFIG_ESP_HOSTED_PRIV_SDIO_PIN_CLK_SLOT_1=53
38 CONFIG_ESP_HOSTED_PRIV_SDIO_PIN_D0_SLOT_1=52
39 CONFIG_ESP_HOSTED_PRIV_SDIO_PIN_D1_4BIT_BUS_SLOT_1=51
40 CONFIG_ESP_HOSTED_PRIV_SDIO_PIN_D2_4BIT_BUS_SLOT_1=50
41 CONFIG_ESP_HOSTED_PRIV_SDIO_PIN_D3_4BIT_BUS_SLOT_1=49
42 CONFIG_ESP_HOSTED_SDIO_PIN_CMD=54
43 CONFIG_ESP_HOSTED_SDIO_PIN_CLK=53
44 CONFIG_ESP_HOSTED_SDIO_PIN_D0=52
45 CONFIG_ESP_HOSTED_SDIO_PIN_D1=51
46 CONFIG_ESP_HOSTED_SDIO_PIN_D2=50
47 CONFIG_ESP_HOSTED_SDIO_PIN_D3=49
48 CONFIG_ESP_HOSTED_SDIO_PRIV_PIN_D1_4BIT_BUS=51
49 CONFIG_ESP_HOSTED_SDIO_RESET_ACTIVE_HIGH=y
50 CONFIG_ESP_HOSTED_SDIO_GPIO_RESET_SLAVE=20
51 CONFIG_ESP_HOSTED_GPIO_SLAVE_RESET_SLAVE=20
52 CONFIG_ESP_HOSTED_SDIO_RESET_DELAY_MS=1500

Executing task: C:\Users\y\j.platformio\penv\Scripts\platformio.exe device monitor --environment crowpanel-p4-50 --port COM5
MS
I (806) H_API: ** add_esp_wifi_remote_channels **
I (811) H_SDIO_DRV: sdio_data_to_rx_buf_task started
I (816) c6-sdio-ota: [PASS] esp_hosted_init OK in 18ms
I (820) c6-sdio-ota: [ID] Connecting to c6 slave...
I (825) H_API: ESP-Hosted Try to communicate with ESP-Hosted slave

I (831) transport: Attempt connection with slave: retry[0]
W (836) H_SDIO_DRV: Reset slave using GPIO[20]
I (840) sdio_wrapper: esp GPIO [20] configured
I (2364) sdio_wrapper: SDIO master Slot 2, Data Linear 4 bit, Freq(0Hz) (100000Hz)
I (2364) sdio_wrapper: GPIOs: CLK[53] CMD[54] D0[52] D1[51] D2[50] D3[49] Slave_Reset[20]
I (2368) sdio_wrapper: Queues: Tx[20] Rx[20] SDIO-Rx-Node[1]
I (2408) sdio_wrapper: Function 0 Blocksize: 512
I (2409) sdio_wrapper: Function 1 Blocksize: 512
I (2509) H_SDIO_DRV: Card init success, TRANSPORT_RX_ACTIVE
I (2509) transport: set_transport_state: 1
I (2509) transport: Waiting for esp_hosted slave to be ready
I (2515) H_SDIO_DRV: SDIO Host operating in STREAMING MODE
I (2515) H_SDIO_DRV: Starting SDIO process rx task
I (2518) H_SDIO_DRV: Open data path at slave
```

Upgraded version: The current P4 host detects that the C6 slave firmware version is already v2.12.3

The firmware to be flashed locally is exactly the same as the existing version of the slave

The system determines that OTA upgrade is not needed and directly terminates the flashing process (HALTED).

```
Executing task: C:\Users\y\j.platformio\penv\Scripts\platformio.exe device monitor --environment crowpanel-p4-50 --port COM5
MS
I (2634) esp_cli: Registering command: task-dump
I (2638) esp_cli: Registering command: cpu-dump
I (2642) esp_cli: Registering command: heap-trace
I (2647) esp_cli: Registering command: sock-dump
I (2651) esp_cli: Registering command: host-power-save
I (2656) vhci_drv: Host BT Support: Enabled
I (2660) vhci_drv: BT Transport Type: VHCI
I (2664) H_SDIO_DRV: Received INIT event
I (2667) H_SDIO_DRV: Event type: 0x22
I (2671) H_SDIO_DRV: Write thread started
I (2712) c6-sdio-ota: [PASS] Connected to c6 slave in 1887ms
W (2712) c6-sdio-ota: [PHASE] 0/5 INIT complete t=2013ms
W (2712) c6-sdio-ota: [PHASE] 1/5 VERSION-QUERY start t=2013ms
I (2816) RPC_WRAP: Coprocessor Boot-up
I (2821) c6-sdio-ota: [ID] C6 firmware: 2.12.3
I (2821) c6-sdio-ota: [PASS] C6 already at v2.12.3 - OTA not needed
W (2821) c6-sdio-ota: [PHASE] 5/5 SUMMARY t=2122ms
W (2825) c6-sdio-ota: RESULT: C6 already at v2.12.3 - no OTA needed
I (2832) c6-sdio-ota: HALTED.
```